Highlights

Investigating the Impact of Label-flipping Attacks against Federated Learning for Collaborative Intrusion Detection[⋆]

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- The first systematic analysis of the impact of label-flipping attacks on Collaborative Intrusion Detection Systems (CIDSs) leveraging Federated Learning (FL), answering a set of research questions.
- A comprehensive understanding of the impact of these attacks on the performance of the learning algorithm.
- A reusable methodology to assess the impact of label-flipping attacks on FL, with experiments that can be easily replicated and extended to other datasets, attack types, and mitigation strategies.
- An automation framework to facilitate the evaluation of FL approaches under different attack scenarios, built upon Flower and Hydra.
- A demonstration that similarity-based mitigation strategies will falter in heterogeneous settings.

Investigating the Impact of Label-flipping Attacks against Federated Learning for Collaborative Intrusion Detection

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Abstract

The recent advances in Federated Learning (FL) and its promise of privacy-preserving information sharing have led to a renewed interest in the development of collaborative models for Intrusion Detection Systems (IDSs). However, its distributed nature makes FL vulnerable to malicious contributions from its participants, including data poisoning attacks. Label-flipping attacks—where the labels of a subset of the training data are flipped—have been overlooked in the context of IDSs that leverage FL primitives. This work contributes to closing this gap by providing a systematic and comprehensive overview of the impact of label-flipping attacks on Federated Intrusion Detection Systems (FIDSs). We show that the effects of such attacks can range from severe to highly mitigated, depending on hyperparameters and dataset characteristics, and that their mitigation is non-trivial in heterogeneous settings. We discuss these findings in the context of existing literature and propose recommendations for the evaluation of FIDSs. Finally, we provide a methodology and tools to extend our findings to other models and datasets, thus enabling the comparable evaluation of existing and future countermeasures.

Keywords: federated learning, intrusion detection, data-poisoning, label-flipping, systematic analysis, quantitative assessment.

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1. Introduction

Securing information systems is becoming increasingly difficult, whether due to the growing complexity of the systems themselves or the sophistication of the attacks they face. Intrusion detection systems are a critical component of the security infrastructure of any organization, offering complementary defense mechanisms to traditional preventive measures. In this context, Federated Learning (FL) has emerged as a promising approach to build collaborative Intrusion Detection Systems (IDSs) that can leverage the collective knowledge of multiple organizations without compromising the confidentiality of their data [1]. Since 2018, numerous works have experimented with various applications of Federated Intrusion Detection System (FIDS), such as Internet of Things (IoT) [2] or Vehicle-to-Everything (V2X) [3]. Multiple surveys on FL for IDSs have also been published, highlighting the community's interest [4–9].

However, the distributed nature of this type of collaborative system introduces new attack vectors, notably related to the quality of the contributions. Among the challenges identified in the literature, the robustness of FL to poisoning attacks has been a recurring concern [10, 11]. These challenges are further exacerbated in the context of IDSs, where the performance of the learning algorithm directly impacts the security of the monitored system.

While a significant amount of literature exist on mitigation strategies against poisoning attacks in FIDSs, few studies focus on understanding and quantifying the impact of these attacks. In addition, most existing work, either in the context of FIDSs or more generally in FL, focuses on cross-device federations, which are less representative of this collaborative setting. In particular, the effects of label-flipping attacks remained overlooked. To the best of our knowledge, no systematic study has been conducted to understand their impact on FL for Collaborative Intrusion Detection Systems (CIDSs), and the lack of comparability in the existing literature makes it difficult to generalize the published findings.

This paper thus aims to fill these gaps by further investigating label-flipping attacks in FIDSs, and understanding the conditions under which these attacks are most effective. We do so by conducting a systematic and quantitative assessment of the impact of label-flipping attacks on FIDSs. These attacks present apparent simplicity, as they only imply changing the labels of a subset of the training data. For attackers, they are consequently relatively easy to implement, even in a *black-boxed* system. However, such basic manipulations can have a significant impact on the trained global model.

Specifically, this study aims to answer the following research questions.

RQ1. Is the behavior of poisoning attacks predictable?

RQ2. Do hyperparameters influence the impact of poisoning attacks?

RQ3. Are IDS backdoors realistic using label-flipping attacks?

RQ4. Is there a critical threshold where label-flipping attacks begin to impact performance?

RQ5. Are existing techniques well-suited to detect label-flipping attacks?

In summary, the contributions of this work are fivefold:

- the first systematic analysis of the impact of label-flipping attacks on CIDSs leveraging FL;
- A comprehensive understanding of the impact of these attacks on the performance of the learning algorithm;
- A reusable methodology to assess the impact of poisoning attacks on FL, with experiments that can be easily replicated and extended to other datasets, attack types, and mitigation strategies;
- An automation framework to facilitate the evaluation of FL approaches under different attack scenarios, built upon Flower [12] and Hydra [13].
- A demonstration that similarity-based mitigation strategies will falter in heterogeneous settings

The content of this article builds upon our work previously published at the International Conference on Availability, Reliability and Security (ARES) [14]. Its remainder is organized as follows. First, Section 2 presents the necessary background information on FL and poisoning attacks in this context. Section 3 details related works, notably the existing analyses on the impact of poisoning attacks on FL. In Section 4, we discuss the methodology used to carry out the experiments, with an emphasis on reproducibility. Then, Section 5 presents the results of the experiments, answering the research questions. Finally, Section 6 discusses the implications of our findings and the limitations of this study. We conclude in Section 7 by summarizing the main takeaways and discussing future work that could be conducted to extend this study.

2. Preliminaries

2.1. Federated Learning for Intrusion Detection

FL is a distributed learning paradigm that enables training a global model without sharing local data [1]. Model training is structured in rounds, where an orchestrating server S randomly tasks n participants $p_i, i \in [\![1, n]\!]$ from a pool of participants P to train a model w_i^r at each round r. The round ends with the aggregation of the collected models into a new global model W_r, which is redistributed to the clients as a starting point for the next round $(r + 1)$. Depending on the scale of the federation—*i.e.*, cross-silo or cross-device [15]—, the fraction C of selected participants per round can vary. The model architecture and hyperparameters are the same across the federation, but each participant owns a local dataset d_i that is not shared with the others.

In the context of IDSs, participants generally seek to classify network flows into two classes (*benign* and *malicious*), which is a binary classification task. Consequently, each dataset d_i is a set of data points representing flows, and it

associates each flow $\vec{x_j}$ with its label $\vec{y_j}$. We refer to the elements of a dataset as samples. The distribution of each dataset d_i depends on the collected traffic, and therefore varies depending on the devices or services active on the network. Various degrees of similarity between clients exist, from purely Independent and Identically Distributed (IID) partitioning to *pathological* non-IID (NIID) settings, where all clients have unique data-distributions without class overlap [16].

To train their model, the participants aim at minimizing a loss function

$$\mathcal{L}(w, \vec{x_j}, y_j), j \in [\![1, |d_i|]\!], \tag{1}$$

where $\vec{x_j}$ and y_j are the sample and its label, respectively. To do so, they use an optimizer leveraging the principles of Stochastic Gradient Descent (SGD). After computing the gradients $\nabla\mathcal{L}(w, \vec{x_j}, y_j)$, they update their model as

$$w_i^{r+1} \leftarrow w - \eta\nabla\mathcal{L}(w_i, d_i), \tag{2}$$

where η is the learning rate. Alternatively, they upload the gradients to the server, which will update the global model W_r as a function of the gradients $\{\nabla\mathcal{L}(w_i, d_i) \mid i \in [\![1, n]\!]\}$ (*e.g.*, `FedSGD` [1]). Whether the model is updated locally or globally, the server aggregates the uploaded parameters and broadcasts the new global model to the participants.

The definitive choice of the aggregation method depends on the learning objective, but also on the assumed data distribution between the participants. The original *pathological*-NIID settings [1] is rarely encountered in practice, as a binary-classification task typically requires both benign and malicious data. Consequently, the literature often relies on one of two strategies [17]: (a) distributing a dataset among clients and then removing samples from n attack classes from each; or (b) distributing the benign data among clients and then assigning samples from n attack classes to each client, with or without class overlap. Campos et al. [9] also examine alternative partitioning strategies that are deemed more realistic, such as separating data by IP addresses. Note that this does not account for resource or network heterogeneity, which are less relevant in an inter-organization, cross-silo federation. However, the literature has shown their importance in cross-device settings, where stragglers and slow networks can significantly affect the training process [15].

2.2. Poisoning Attacks in Federated Learning

The attack surface of FL is rather broad, as it includes various types of threats, such as poisoning and privacy attacks [18]. Authors often refer to poisoning attacks in FL as *Byzantine* attacks, since they are analogous to the Byzantine General problem [19] in distributed systems. Likewise, the term *Sybil attacks* [20] is frequently used to refer to the problem of *colluding attackers* [21].

Poisoning attacks can be classified into two main categories depending on the phase in which they are perpetrated: model-poisoning [10] or data-poisoning [11]. Model-poisoning attacks aim at manipulating the model parameters, usually

during or after training, to deviate the aggregated model from the global optimum [22]. Data-poisoning attacks, on the other hand, occur before the training phase and manipulate data samples to degrade performance, cause misclassification, or introduce backdoors [18].

Data poisoning attacks can be categorized into clean-label and label-flipping attacks. Clean-label attacks manipulate the samples to be misclassified, either by adding new samples [23] or by modifying existing ones [24]. Label-flipping attacks, on the other hand, change the labels of the samples by flipping them to a different class [11].

In addition, most poisoning attacks can be further separated into *untargeted* and *targeted* attacks. Untargeted attacks randomly select samples to be manipulated and are usually easier to detect as they have a higher impact on the model's performance. Targeted attacks, on the other hand, select samples based on a specific criterion, such as the class to be targeted. In a CIDS context, targeted attacks can be used to introduce backdoors—*i.e.*, making a specific attack class be misclassified as benign—or cause targeted misclassification.

2.3. Countermeasures

Algorithmic solutions exist to mitigate these threats by using more robust aggregation strategies. This includes Krum [25], which elects the model update that is closest to the others as the new global model, or Trimmed Mean [26], which aggregates the model updates by trimming the highest and lowest values. Both methods share the goal of filtering out outliers, assuming that they would negatively impact the global model.

This notion of distance between updates is often leveraged to different extents in the literature. Clustering-based methods [27, 28] group similar updates together, either to eliminate outliers or to identify clusters of updates that may be indicative of an attack. Similarly, other similarity-analysis techniques [21, 29] use the distance between updates to flag participants that might be suspiciously similar.

Although other less conventional methods exist, such as client-side evaluation [30, 31], they are often more computationally expensive and less scalable. More generally, the trade-off between computational overhead and robustness is a recurring theme in the literature, and the design of a countermeasure often depends on the specific threat model and requires a deep understanding of the attack capabilities.

3. Related Work

The literature on the impact of poisoning attacks on FL [10, 11, 32, 33] provides insight into the behavior of poisoning attacks on generic Machine Learning (ML) tasks, such as image classification or natural language processing. Nuding and Mayer [32] focus specifically on backdoor attacks, and emphasize the importance of the choice of the trigger pattern. Fang et al. [22], Sun et al. [33] rather study model-poisoning attacks. While often more effective than data-poisoning

attacks, model-poisoning attacks are more complex to implement, as they require access to the uploaded models and knowledge of their functioning. The work of Tolpegin et al. [11] is closest to ours, as it focuses only on label-flipping attacks. Among the most notable outcomes, the authors exhibit that targeted attacks are especially effective, having small to no impact outside the targeted class. We review notably review the applicability of these conclusions to the context of IDSs in Section 5.3.

In the context of IDSs, the literature on the impact of poisoning attacks on FL is scarcer. Zhang et al. [23] provide a systematic analysis of clean-label data-poisoning attacks, where they use Generative Adversarial Networkss (GANs) to generate poisoned samples. Other works discuss clean-label attacks to a lesser extent [34, 35]. These works offer a complementary view to our study, although the lack of comparability of the datasets and models used makes it difficult to draw direct comparisons.

Meanwhile, Merzouk et al. [24] provide a comprehensive analysis on ML backdoors in FL for IDSs. ML backdoors work by manipulating samples to associate a specific trigger pattern with a given class so that the model misclassifies samples containing the trigger pattern. We compare its efficiency with label-flipping attacks in Section 5.3. Finally, Yang et al. [36] discuss the specific aspects of label-flipping attacks in the context of FL for IDSs, using two different datasets, NSL-KDD [37] and UNSW-NB15 [38]. However, they only implement label-flipping as a random selection of malicious samples to be flipped, which makes the results less comparable.

4. Methodology

Assessing the impact of data-poisoning over FL implies reviewing a consequent amount of parameters and configurations. To optimize our work and make it easily reproducible, the results presented in Section 5 have been generated using a purposely designed evaluation framework based on Flower [12] and Hydra [13]. We follow the ACM guidelines and terminology [39], and take measures to ensure the *reusability* of our artifacts, the *reproducibility* of our results, and the *replicability* of our experiments. Specifically:

1. We provide the methodology and all parameters necessary to reimplement and replicate the experiments;
2. Dependencies are pinned using Poetry for Python and Nix for system, allowing the entire software pipeline to be executed in the same conditions;
3. All experiments are seeded where possible, which makes the results reproducible within a three decimal precision[2];
4. The results and the code to generate them are available in open access[3],

[2]The distributed nature of FL makes it difficult to ensure the complete reproducibility of the results.

[3]`https://github.com/leolavaur/comsec_2024`

Table 1: Class distribution of CIC-CSE-IDS2018 (left) and UNSW-NB15 (right) [40, 41].

Class	Sampled	Full
Benign	880,623	16,635,567
Malicious	119,377	2,258,141
· DDoS	73,558	1,390,270
· DoS	25,574	483,999
· Bot	7,595	143,097
· Brute Force	6,525	123,982
· Infiltration	6,108	116,361
· Injection	17	432
Total	1,000,000	18,893,708

Class	Sampled	Full
Benign	960,078	2,295,222
Malicious	39,922	95,053
· Exploits	13,187	31,551
· Fuzzers	9,377	22,310
· Generic	6,976	16,560
· Reconnaissance	5,352	12,779
· DoS	2,455	5,794
· Analysis	969	2,299
· Backdoor	925	2,169
· Shellcode	617	1,427
· Worms	64	164
Total	1,000,000	2,390,275

as are the datasets[4].

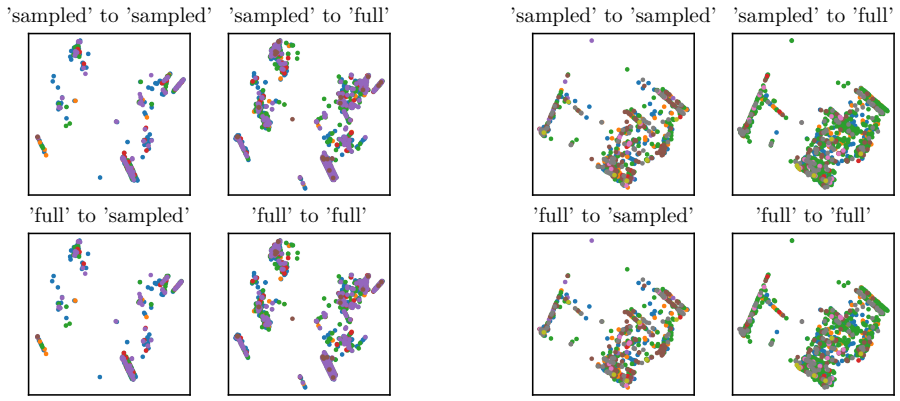
The results presented in this paper amount to 10 670 unique runs, and close to 1 613 cumulated computing hours on two NixOS servers with 96 cores, 768 GB of RAM and 2 Nvidia Tesla T4 each.

4.1. Dataset and Pre-processing

Due to the scale of the required experiments, we require datasets that are both representative of the problem and small enough to be processed in a reasonable amount of time. Recent works on FL and IDS [42] proposed a standardized feature set (NF-V2) making cross-dataset FL setups easier, which uses nProbe [43] and its NetFlow V9 format to extract features. The authors notably provide converted versions of known datasets, including CSE-CIC-IDS2018 [44] and UNSW-NB15 [38], the two most used generic Network-based Intrusion Detection System (NIDS) datasets in the literature. UNSW-NB15 is a dataset generated using the IXIA PerfectStorm tool to simulate a real network environment and contains 9 attack classes: *Analysis*, *Backdoor*, *DoS*, *Exploits*, *Fuzzers*, *Generic*, *Reconnaissance*, *Shellcode*, and *Worms*. CSE-CIC-IDS2018 is a larger scale version of the CIC-IDS2017 [44] dataset, generated using AWS. It contains 14 attacks labels grouped in 6 classes: *DDoS*, *DoS*, *Bot*, *Brute Force*, *Infiltration*, and *Injection*.

We use the "sampled" version of the datasets provided by the authors [41], to make our experiments more manageable. Table 1 summarizes the distribution of the classes in the datasets. After preprocessing, we evenly split the dataset for the experiments, ensuring the same class distribution in the training and testing sets. 80% of the dataset is used for training, and 20% for testing. We purposely do not use a proper validation set, as the goal is to measure the performance delta related to the impact of the attacks on the global model, and not to optimize the model's hyperparameters. Indeed, the parameters are kept constant throughout the experiments, as detailed in Table 2. We sometimes

[4] https://staff.itee.uq.edu.au/marius/NIDS_datasets/



(a) CIC-CSE-IDS2018 (b) UNSW-NB15

Figure 1: Cross-projections of the malicious traffic from the two datasets in 2D using Principal Component Analysis (PCA). On top, the frame of reference is computed using the `sampled` dataset, and on the bottom the `full` dataset. The `sampled` dataset is then projected on the left, the `full` dataset on the right.

refer to the datasets as `cicids` and `nb15` for brevity in the remaining of this paper.

To assess the representativity of the datasets sampling, we compare the projections in two dimensions of the two datasets using PCA. Figure 1 presents cross-projections results, depending on the datasets used to generate the projection frame. There are consequent overlaps between the classes in this projection, implying that either 2 dimensions are not enough to separate the classes, or there are features that are not relevant to the classification task. Yet, the projected patterns are identical between the two datasets, which indicates that the sampling process does not introduce significant distribution bias in the dataset. Therefore, the experiments performed on the sampled datasets should be representative of the observed behaviors in the original dataset.

4.2. Local and Federated Training

We use a simple Multilayer Perceptron (MLP) model with two hidden layers, as implemented by Popoola et al. [45] who use the same datasets; a summary of the model's parameters is available in Table 2. Trained centrally, this model reaches an F1-score of 0.966 and an accuracy of 0.992 on our sampled CIC-CSE-IDS2018, and 0.945 and 0.995 on the UNSW-NB15 dataset, respectively. These values can be considered as baselines for the FL experiments.

We focus on the impact of data-poisoning specifically, and therefore omit other factors that could hurt the performance of the model, such as client heterogeneity or disconnections. We also specifically concentrate our efforts on a collaborative cross-silo setting, where all clients are available at each round and $C = 1$. In such setting, scalability is less of a concern, but the quality of the participants' data cannot be guaranteed. Consequently, the dataset is partitioned

Table 2: Learning hyperparameters.

Hyperparameter	Value
Learning rate	0.0001
Hidden layers activation	ReLU
Output layer activation	Sigmoid
Input shape	49
Number of hidden layers	2
Size of the hidden layers	128
Optimizer	Adam
Loss function	Binary cross-entropy
Aggregation	`FedAvg`

into 10 IID shards of 80,000 data points, and each client is assigned with one shard. On the server, the uploaded models are aggregated using `FedAvg`—which, since the local datasets are of similar sizes, is equivalent to a simple average of the weights.

4.3. Attack Model and Implementation

We consider data-poisoning attacks where malicious participants can alter their local datasets before training. This definition covers both, participants that have been compromised and those that are deliberately modifying their data. Further, this scenario will always be available, even with a secure and immutable FL client software. Specifically, we implement data-poisoning using label-flipping attacks, where the attacker changes the label y of a sample to a new label y_p; *i.e.*, $y_p = \neg y$ in a binary-classification problem.

Attacker's Objective. We consider two types of objectives for the attacker depending on the type of attack leveraged. With *targeted* attacks, the attacker aims to make a specific attack pattern undetectable, and therefore act as a backdoor in the IDS. This is implemented by labeling a randomly selected fraction of a specific attack class (*e.g.*, *DDoS*) as benign. With *untargeted* attacks, on the other hand, his goal is to produce high =False Positives Rate (FPR) and False Negative Rate (FNR), which can overwhelm the human operators investigating the alerts. Here, a random fraction of the entire dataset is altered, where the label of each sample is flipped from benign to attack and vice versa. The proportion of samples that are altered is controlled by the Data Poisoning Rate (DPR), which is the ratio of samples matching the target that are altered by each attacker on a specific round. We note the DPR, or *local poisoning rate*, as α.

Attacker's Knowledge and Capabilities. We consider attackers to be *gray-box* adversaries, *i.e.*, they have the same knowledge as benign clients, but are unable to modify the system's behavior, neither locally nor on the server. Further, we consider that multiple attackers can be present in the system, and that they can act in concert. This scenario is referred to as *colluding attackers*. In this case,

Table 3: Experimental parameters. Default parameters are highlighted in bold and are used if not specified otherwise. As targets are dataset-specific, the available targets are listed for each dataset.

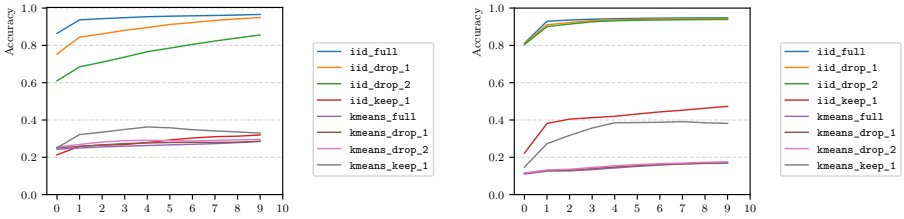
Parameter	Values
dataset	cicids, nb15
batch_size	32, 128, **512**
epochs	**100_10x10**, 100_4x25, 100_1x100, 300_10x30, 300_4x75, 300_1x300
distribution	10-0, 9-1, 7-3, **5-5**, 3-7
scenario	continuous-{10,30,60,70,80,90,95,99}, **continuous-100**, late-3, redemption-3
target (cicids)	**untargeted**, bot, dos, ddos, bruteforce, infiltration, injection
target (nb15)	**untargeted**, generic, analysis, worms, backdoor, exploits, untargeted, shellcode, dos, reconnaissance, fuzzers
partitioner	iid_drop_1, iid_drop_2, **iid_full**, iid_keep_1, kmeans_drop_1, kmeans_drop_2, kmeans_full, kmeans_keep_1
seed	1313, 1977, 327, 5555, 501, 421, 3263827, 2187, 1138, 6567

the attackers share the same target and DPR. The proportion of attackers can vary from one single malicious client to a majority of them being malicious, and is expressed as τ, or Model Poisoning Rate (MPR) [24]. Note that in the context of IID partitioning, the overall poisoning rate could be regarded as $\alpha \times \tau$. This simplification is however not accurate in other partitioning strategies.

4.4. Experiments

We design a set of experiments to answer the research questions outlined in Section 1. All experiments share a common set of constants which are complemented by a set of variable parameters. Table 3 summaries the parameters available for the experiments. Each combination is tested 10 times using a set of 10 different seeds to study the predictability of the results. Specifically, the seed impacts data-partitioning operations (both between training and testing sets and between clients afterward), sample selection in poisoning attacks, and the random weights of the initial model. It also impacts all the random operations (such as data shuffling) performed during model training.

The `epochs` parameter controls the aggregation frequency, *i.e.*, the number of local epochs per round $\mathcal{E}$, as well as the number of rounds R. Logically, `batch_size` controls the size of the batches used at each iteration during training. The global number of local epochs per client is kept to 100 or 300 to preserve comparability. The `distribution` represents the number of legitimate and malicious clients in the system, and consequently the proportion of attackers. The key `scenario` represents the behavior of the attackers. The scenarios defined as `continuous-`α represent a constant poisoning rate of α throughout the training process. The scenarios named `late-`r and `redemption-`r produce an attack with $\alpha = 100$ that starts or ends at round r, respectively. The `target`



(a) CIC-CSE-IDS2018 (b) UNSW-NB15

Figure 2: Impact of the partitioning strategy on the model's performance.

parameter represents the target of the attack as defined in Section 4.3; each attack class is available as a target.

4.4.1. Partitioning Strategies

A particular parameter in this selection is the `partitioner`, which defines the strategy used to partition the dataset among the clients. As mentioned in Section 2.1, the literature of FIDSs usually considers either IID partitioning or basic NIID scenarios where clients randomly drop attack classes. To study the usage of similarity metrics to detect poisoning attacks, we consider eight different partitioning strategies, with different levels of heterogeneity that range from IID partitioning to *pathological*[5] settings without distribution overlap.

The `partitioners` marked named `iid_*` distribute the data in an IID manner, where each client receives a stratified[6] subset of the dataset. Then, `full`, `drop_*`, and `keep_*` refer to the way the attack classes are processed: all kept, certain classes dropped, or certain classes kept, respectively. The `kmeans_*` partitioners use a k-means clustering algorithm to partition the benign traffic among the clients, and then distribute the attack classes based on their attribute (`full`, `drop_*`, `keep_*`). Because k-means algorithm minimizes the intra-cluster variance, it increases the heterogeneity of the clients' datasets. Figure 2 illustrates the performance of the model under different partitioning strategies.

4.5. Metrics

To quantify how the experiment parameters impact the global model, we define a set of metrics to measure the Attack Success Rate (ASR) of poisoning attacks. The definition of the ASR differs depending on the type of attack, according to the attacker's objective defined in Section 4.3. Because ASR is based on performance and there is no perfect model, we distinguish the Absolute Attack Success Rate (AASR) measured in the attack scenario, from the Relative

[5]In *pathological* NIID settings, each client owns a unique class without any overlap with the others. Note that these settings are particularly unrealistic for IDS purposes.

[6]Stratified sampling ensures that the class distribution remains the same among the different shards. It is usually done using the labels of the samples, but any categorical feature can be used.

Attack Success Rate (RASR) which also considers the nominal performance without attacks. Formally, the RASR is defined as:

$$\text{RASR} = \frac{\max(\text{AASR}_{benign}, \text{AASR}_{attack}) - \text{AASR}_{benign}}{1 - \text{AASR}_{benign}}, \tag{3}$$

where AASR_{benign} and AASR_{attack} are the AASR of the *benign* and *attack* scenarios respectively, under the same set of parameters. This is made possible thanks to the framework's reproducibility, which ensures that two experiments started with the same seed will run under the same conditions. Following the definitions in Section 4.3, we then defined two variations of the AASR depending on the attacker's objective. Both are computed based on the confusion matrix of the model: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).

Targeted attacks: Malicious participants leverage targeted attacks to make a specific attack pattern undetectable. Therefore, a successful attack forces classification of the relevant attack samples as benign. The AASR is then defined as the miss rate of the targeted attack, *i.e.*

$$\text{AASR} = \frac{\text{FN}_c}{\text{TP}_c + \text{FN}_c}, \tag{4}$$

where c is a specific attack class of the dataset.

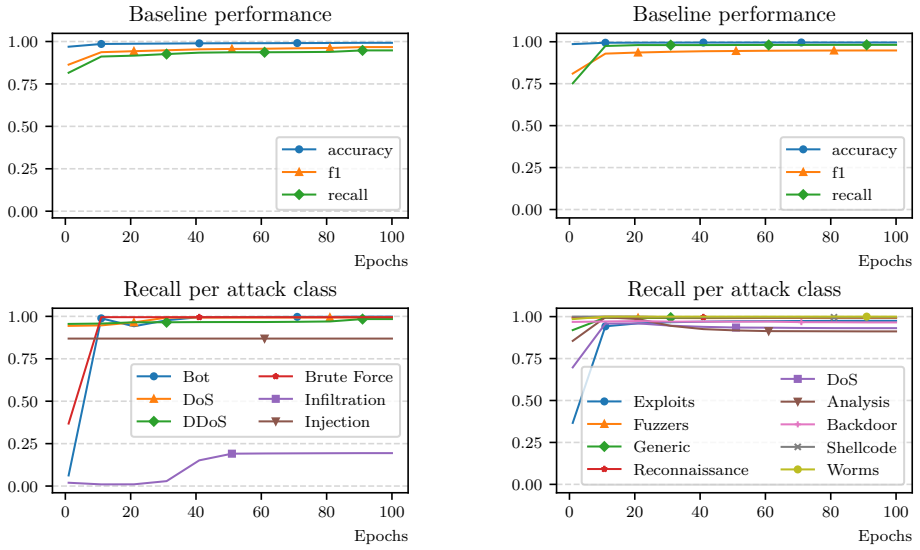
Untargeted Attacks: Untargeted attacks aim at degrading the overall classification rate of the model. Consequently, the AASR is defined as the misclassification rate of the model, *i.e.*

$$\text{AASR} = \frac{\text{FP} + \text{FN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = 1 - \text{Accuracy}. \tag{5}$$

Additionally, we use traditional binary classification metrics to observe the performance of the model under various conditions, as identified in existing surveys [5, 9] These metrics include *accuracy*, *F1-score*, and *miss rate*. Notably, we consider the Main Task Accuracy (MTA), defined as the accuracy of the benign clients obtained on the testing set, to measure the impact of the attacks on the model's nominal performance. All metrics are aggregated over the 10 runs of each experiment and the mean and standard deviation are reported for the selected metric.

5. Results

The results presented in this section aim to answer the research questions defined in Section 1. Figure 3 presents the performance of the global model without malicious participants to serve as a comparison baseline. The model displays relatively high performance on both datasets, *i.e.*, above 0.9 for accuracy, recall, and F1-score. However, some classes are more challenging to detect



(a) CIC-CSE-IDS2018.

(b) UNSW-NB15.

Figure 3: Baseline performance of the global model without malicious participants. The accuracy, F1-score, and recall illustrate the performance that can be expected from the global model under the conditions selected for this study ($\mathcal{E} = 10$, $\beta = 512$). The recall of the available attack classes shall serve as a reference for the RASR of targeted attacks.

Table 4: Experiment parameters for RQ1

Is the behavior of poisoning attacks predictable?	
batch_size	32, 512
epochs	300_10x30, 300_4x75, 300_1x300
distribution	5-5
dataset	cicids, nb15

than others. The low representation of the "Injection" class in `cicids` (around 0.0017%, see Table 1) prevents the model from learning from it, which results in no significant improvement over time (see Figure 3a). The "Infiltration" class is more represented in the dataset (0.6108%, approximately the same as the "Brute Force" and "Bot" classes), but remains difficult to learn due to its apparent similarity to benign traffic. Consequently, it never exceeds a recall of 0.2. The detection in `nb15` is better overall, with the exception of the "Analysis" and "DoS" classes that score below 0.9, although they are not the least represented classes in the dataset.

5.1. Impact Predictability

A preliminary question to answer before quantifying the effects of label-flipping is whether the behavior of poisoning attacks is predictable. This is

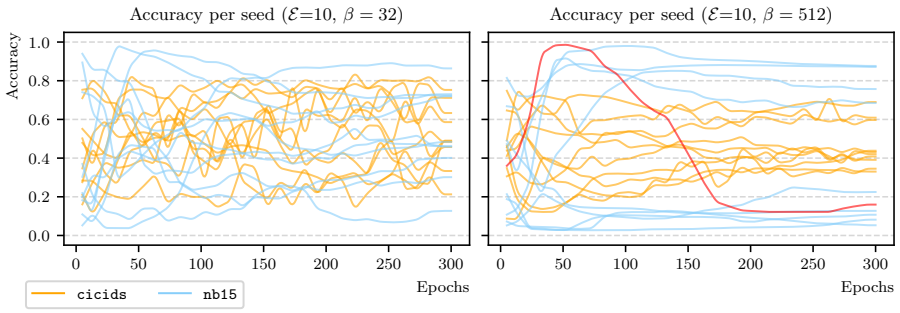
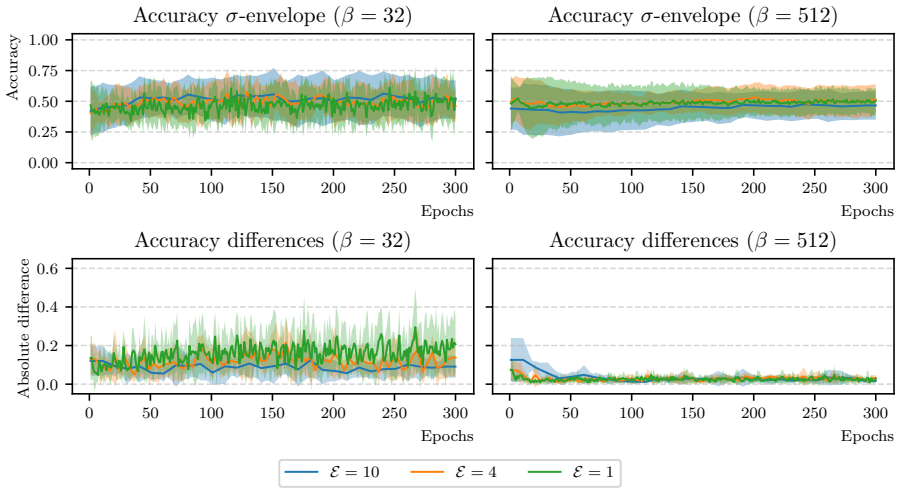


Figure 4: Impact of the dataset on the accuracy under poisoning. $\tau = 0.5, \mathcal{E} = 10$. The x-axis represents the number of local epochs. Each line represents the accuracy over time for a specific seed. The seed 6567 on `nb15` ($\beta = 512$) is depicted in red.

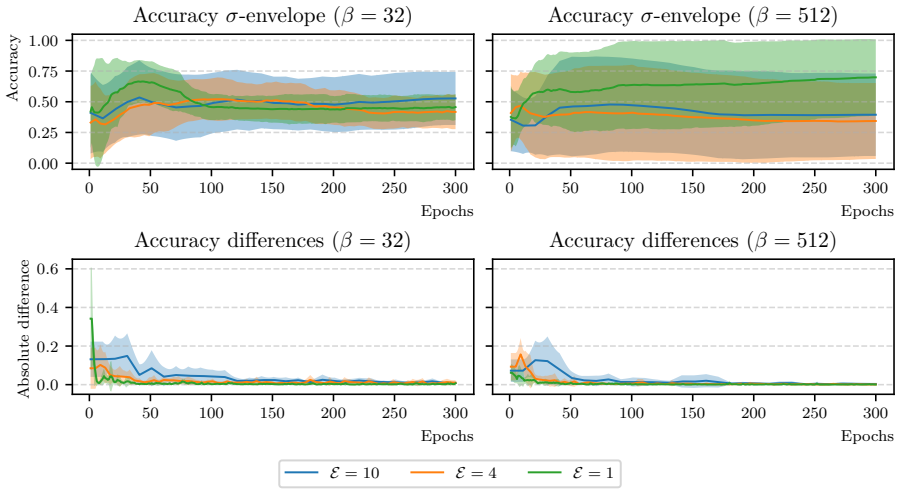
a requirement for generalizing our results to other datasets and models, and comparing the findings with current and future studies. Due to computation constraints, we focus in this part on the parameters that have the most significant impact on the results, as we need to perform longer experiments to ensure the stability of the results. Specifically, the selected distribution contains 50% of malicious participants (*i.e.*, $\tau = 0.5$), which roughly equates to 50% of the training data being poisoned. The experiments are carried out over 300 epochs, with three different aggregation frequencies ($\mathcal{E} \in \{1, 4, 10\}$) and two different batch sizes ($\beta \in \{32, 512\}$). Table 4 summarizes the parameters used for this experiment.

Figure 4 illustrates the accuracy of the global model over time for each seed. This figure brings two key insights. First, the different runs exhibit consequent variability, especially in the early stages of training. This is true for the runs themselves, but also for between runs on the same datasets. Using the same parameters, the accuracy of the global model varies from 0.2 to 0.8 after 100 epochs on `cicids` (with $\beta = 32$) and from 0.05 to 0.95 on `nb15`. Some runs present surprisingly volatile results, such as the seed 6567 on `nb15` ($\beta = 512$), depicted in red in Figure 4, which starts below 0.4, then increases to close to 1.0, before dropping below 0.2 and stabilizing. Overall, `nb15` presents a higher variability than `cicids`, but with runs that are more stable.

The dispersion decreases with time, given a large enough batch size, as shown in Figure 5. More importantly, each run progressively converges to a steadier state. The absolute accuracy differences in the bottom rows of Figure 5 indeed decrease over the first epochs, before plateauing. It can be interpreted as a consequence of the complexity of the learning tasks, which becomes harder as clients possess different labels for similar samples. Therefore, the problem probably admits a high number of local minima, which are reached depending on the seed. In contrast, the difference between rounds using $\mathcal{E} = 32$ on `cicids` (see Figure 5a) tends to increase over time, illustrating the difficulty for each run to converge to a stable state.



(a) CIC-CSE-IDS2018.



(b) UNSW-NB15.

Figure 5: Studying attack impact predictability over time, with 50% attackers. The x-axis represents the number of local epochs. For each subfigure (Figures 5a and 5b), the top row illustrates each seed's accuracy over time with its standard deviation. The bottom row displays the mean absolute difference ($|\text{acc}_r - \text{acc}_{r-1}|$) over time.

Table 5: Experiment parameters for RQ2

Do hyperparameters influence the impact of poisoning attacks?	
batch_size	32, 128, 512
epochs	100_10x10, 100_4x25, 100_1x100
distribution	10-0, 5-5
scenario	continuous-100, late-3, redemption-3

Answering RQ1 / *Is the behavior of poisoning attacks predictable?*

The behavior of poisoning attacks is not predictable, as the dispersion between results is too important, although only the seed varies. However, the dispersion decreases over time in some conditions, and each run tends to converge to a stable state. *In practice, this makes the impact difficult to predict for a specific attack instance, even though general tendencies can be extrapolated.*

5.2. Hyperparameters Impact

To understand the impact of hyperparameters on the behavior of poisoning attacks, we study the impact of different batch sizes (β) and aggregation frequencies ($\mathcal{E}$). We set again the conditions from Section 5.1 but limit the number of epochs to 100, as most scenarios do not show significant changes after this point (see Section 5.1). Additionally, we evaluate the hyperparameters in the *late* poisoning scenario, where the attackers only start after a 3-round bootstrap period, and in the *redemption* scenario, where the attackers stop after 3 rounds. The experiment parameters are summarized in Table 5.

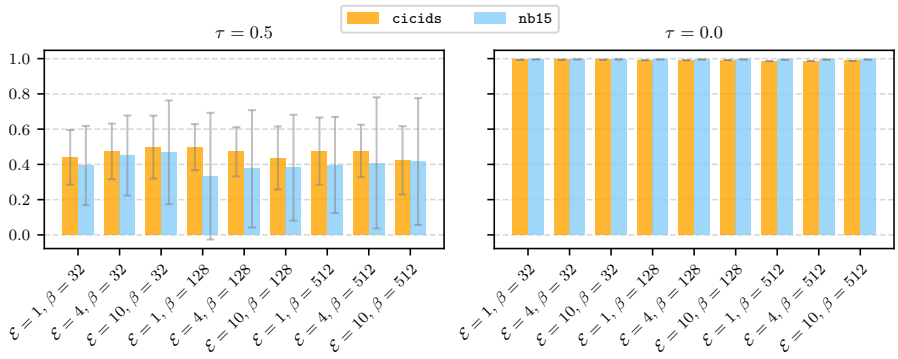
Figure 6 illustrates the influence of hyperparameters on the impact of poisoning attacks. The differences (the two bottom rows) are shown on a bi-symmetric logarithmic scale [46], defined as

$$x' \mapsto \operatorname{sgn}(x) \cdot \log_{10}\left(1 + |\frac{x}{10^{-4}}|\right), \tag{6}$$

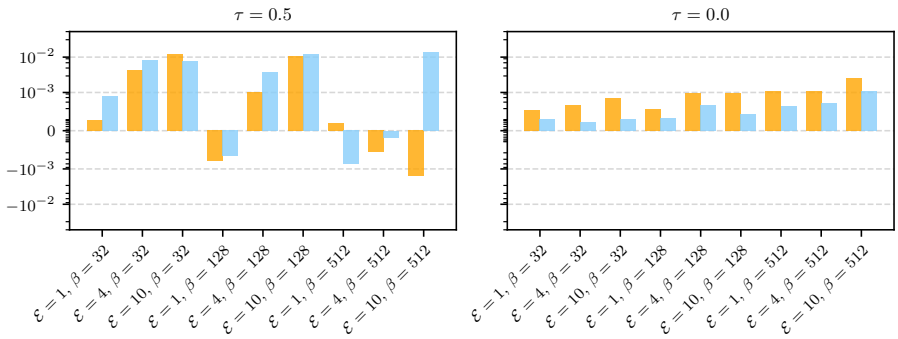
to make up for the consequent differences in scale between combinations. In addition to the accuracy of each parameter combination, Figure 6 also presents the average change in standard deviation, or

$$\frac{1}{R-1}\sum_{r=2}^{R}\sigma^{r} - \sigma^{r-1}, \tag{7}$$

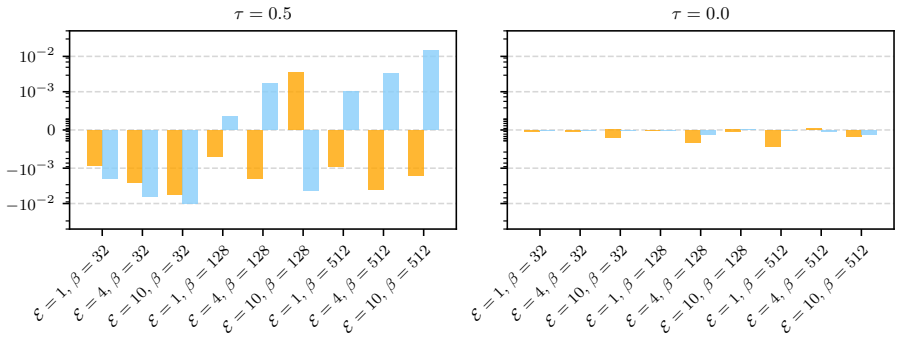
where R is the number of rounds and σ^r is the standard deviation of the accuracy at round r between the different seeds. The average change accuracy is also displayed. For these metrics, a positive value indicates an increase in the observed metric over time.



(a) Mean accuracy

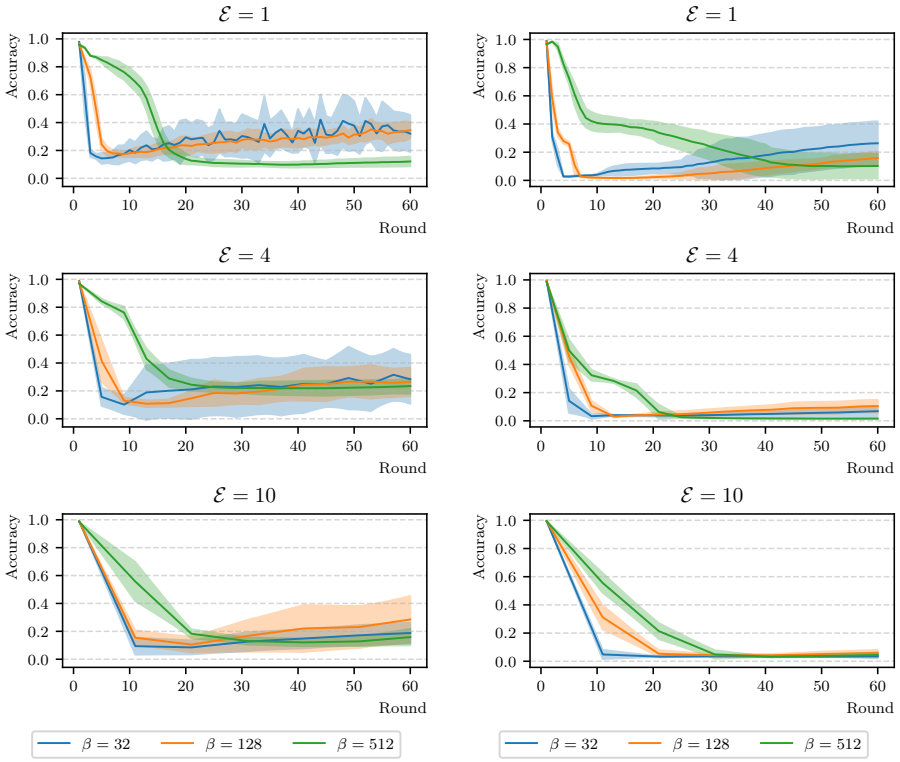


(b) Mean accuracy difference ($x_r - x_{r-1}$)



(c) Mean σ difference ($\sigma_r - \sigma_{r-1}$)

Figure 6: Comparing the influence of hyperparameters on the impact of label-flipping attacks ($\tau \in \{0, 0.5\}$ and $\alpha = 1$). The x-axis represents the number of local epochs. The top row illustrates the mean accuracy for each combination of hyperparameters (β and $\mathcal{E}$). The middle row displays the mean difference in standard deviation (σ) between rounds. The bottom row shows the mean difference in terms of accuracy between rounds.



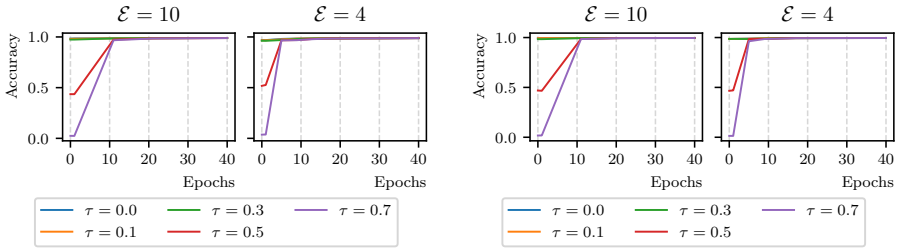
(a) CIC-CSE-IDS2018.

(b) UNSW-NB15.

Figure 7: Impact of hyperparameters on the accuracy of the global model under the `late` scenario. The data is aligned to start at the last benign round before the attack, and the impact is measured over the next 60 epochs (*i.e.*, 6, 24 or 60 rounds depending on the aggregation frequency).

In the continuous scenario studied in Figure 6 (with $\tau = 0.5$ and $\alpha = 1$), the hyperparameters have little impact on the accuracy of the global model. However, $\beta = 32$ presents a slight increase in accuracy over time (Figure 6b) and a decrease in dispersion (Figure 6c). Note that there is close to no dispersion in the accuracy of the benign scenario, as depicted in Figure 6c, confirming that the attack is indeed responsible for the dispersion observed in Section 5.1. Interestingly, the correlations between the two tested datasets are more pronounced with smaller batch sizes, as shown in Figures 6b and 6c where the results become less correlated with $\beta = 512$.

The other hyperparameters do not show any significant correlation. In the end, all tested parameters lead to an accuracy between 0.4 and 0.5 under poisoning, while they all exceed 0.95 without poisoning. This is critically low for intrusion detection, as 0.5 is the score of a random classifier on a balanced binary-classification task. *Tossing a coin would yield better results.*



(a) CIC-CSE-IDS2018.

(b) UNSW-NB15.

Figure 8: Accuracy of the global model after a label-flipping attack. The data is aligned to start at the last epochs before the attack ends, and the impact is measured over the next 40 epochs (*i.e.*, 4 or 10 rounds depending on $\mathcal{E}$).

However, when clients have been given the time to converge before the attack, the impact of hyperparameters becomes more visible, particularly for the batch size, as depicted in Figure 7. While the impact is instantaneous when $\beta = 32$, it takes around 20 epochs with $\beta = 512$ to reach the same accuracy. The dispersion of the results is significantly lower in the latter, as is the reached accuracy, which goes down to 0.25 after 60 epochs. A bigger batch size thus leads to a greater inertia and a lower dispersion of the results when the attack starts, but also to a lower accuracy afterward.

Similar results could be expected in the *redemption* scenario, where the attack stops after 3 rounds. However, Figure 8 presents a quasi-instantaneous recovery of the global model's accuracy after the attack ends. This is expected, as Figure 3 indicates that the accuracy of the global model already exceeds 0.95 at the first round, despite the randomly initialized model parameters provided by the server before the first round. This is also consistent with the results of Zhang et al. [23] on NSL-KDD [37] and UNSW-NB15 [38].

Answering RQ2 / *Do hyperparameters influence the impact of poisoning attacks?*

Although hyperparameters have an impact on the poisoning effect, no combination prevents it: *on average, the performance remains the same.* The dispersion of the results can vary significantly depending on the combination of parameters, especially when the attack occurs after the clients have converged. Then, a smaller batch size leads to a swifter effect, while a larger batch size leads to a greater ASR. In performance-constrained use cases (such as the IoT) in particular, history-based defense mechanisms might need to react faster to mitigate the attack's impact. However, attacks have limited persistence, as the global model's accuracy recovers almost instantaneously. *Consequently, strategies should account for the dynamic behavior of attacks: swift reaction is needed to deal with their almost immediate impact, while offering redemption opportunities.*

Table 6: Experiment parameters for RQ3

Are IDS backdoors realistic using label-flipping attacks?	
distribution	`10-0, 7-3, 5-5, 3-7`
target	`dos, ddos, bot, infiltration, injection`
scenario	`continuous-100`
dataset	`cicids, nb15`

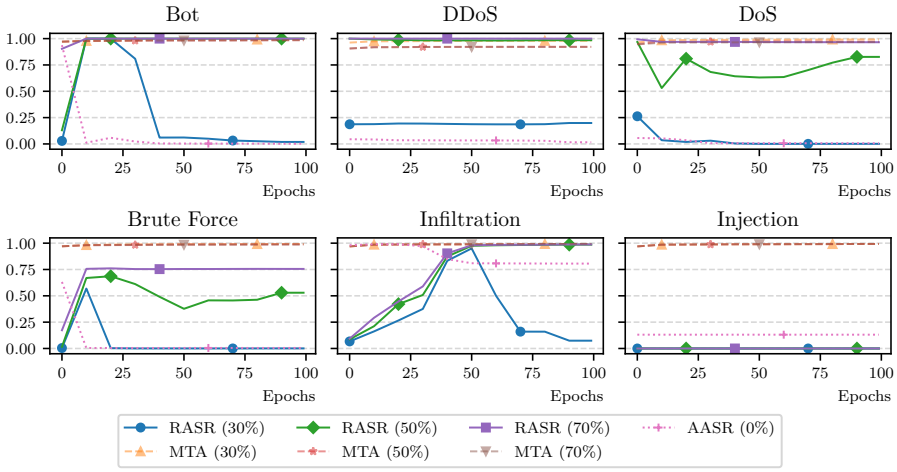
5.3. IDS Backdoors using Label-flipping

One of the main concerns with poisoning attacks is the perspective of backdoors in the IDS, allowing attackers to bypass the system's detection capabilities afterward. To assess this risk, we study the impact of label-flipping attacks with different targets. We consider $\alpha = 100\%$ and various values of τ to assess whether a ASR of 1.0 can be achieved. Table 6 summarizes the parameters used for this experiment.

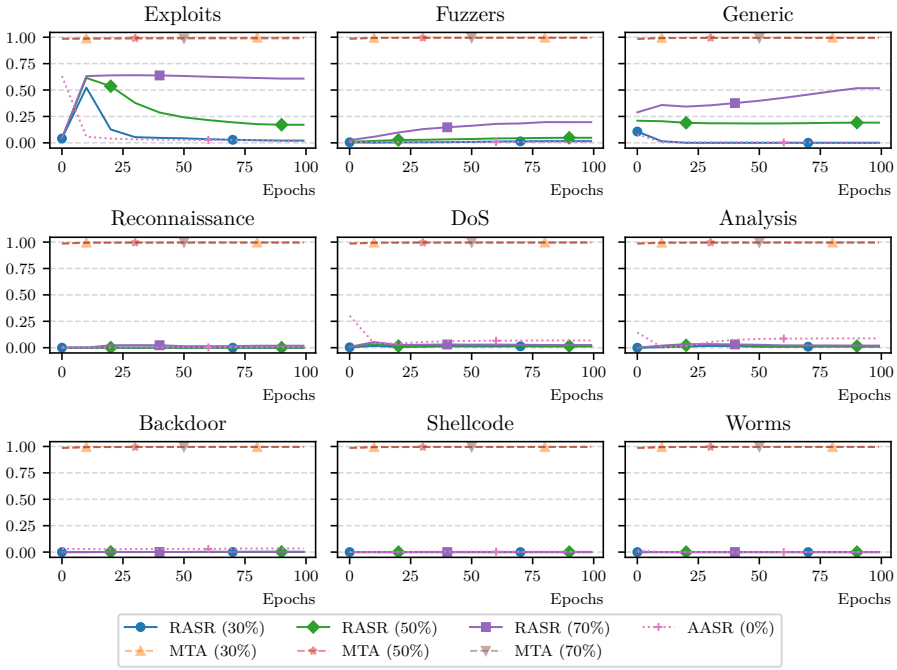
Figure 9 presents the impact of label-flipping attacks on the accuracy of the global model for different attack targets. The figures show the RASR of targeted attacks towards the different classes available in each dataset. To measure the impact of the attack on the overall performance of the model, we also measure the MTA under the different attacks. The AASR of the benign scenario is provided as a reference (*cf*. Figure 3).

The first striking observation is that the results differ greatly between the two datasets. While on `cicids`, some attacks can reach an ASR close to 1.0 given enough attackers, most results on `nb15` remain below 0.25. Some classes seem completely immune to the attack, such as "Backdoor", "Shellcode", and "Worms". Even the most successful attacks on `nb15` barely exceed 0.5. On `cicids`, on the other hand, the results are more convincing. Although 30% of the attackers are not enough to permanently impact the global model's accuracy, multiple classes can reach a RASR close to 1.0 at least momentarily. With $\tau = 0.5$, the RASR of "Bot", "DDoS", and "Infiltration" classes permanently reach 1.0 after a few rounds. Most importantly, the MTA of the global model remains close to its nominal performance, even when the RASR of the targeted classes reaches 1.0, except for the "DDoS" class. Indeed, the "DDoS" class is the most represented in the dataset, with 5.29% of the samples. Therefore, the misclassification of roughly 70% of the samples of this class leads to a more significant impact on the accuracy of the global model.

The result disparity between classes, and more importantly between datasets, is difficult to interpret, although multiple hypotheses can be formulated. First, some classes are significantly less represented in the datasets such as "Injection" in `cicids` or "Analysis", "Backdoor", "Shellcode", and "Worms" in `nb15` (see Table 1). All these targets yielded a RASR close to 0.0. Second, as we consider binary classification tasks, any overlap between the classes' characteristics can be used by the model to infer the correct associations using samples from unaffected classes. For instance, the "Brute Force" and "DoS" classes in `cicids` both



(a) CIC-CSE-IDS2018.



(b) UNSW-NB15.

Figure 9: RASR of targeted label-flipping attacks over time, with $\beta = 512$, $\mathcal{E} = 10$, and $\alpha = 1.0$. The x-axis represents the number of local epochs. The AASR of the benign scenario is provided as a reference for each targeted class.

Table 7: Experiment parameters for RQ4

Is there a critical threshold where label-flipping attacks begin to impact performance?	
distribution	`10-0, 9-1, 7-3, 5-5, 3-7`
scenario	`continuous-{10,30,60,70,80,90,95,99,100}`
target	`untargeted, dos, ddos, bot, infiltration`
dataset	`cicids, nb15`

imply a high number of connections from a single host, and both obtain subpar results when compared to the most efficient attacks.

Answering RQ3 / *Are IDS backdoors realistic using label-flipping attacks?*

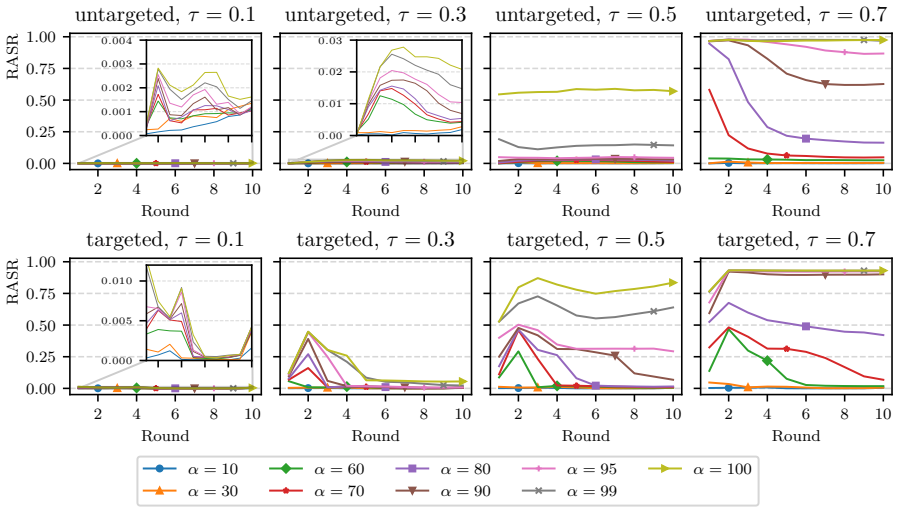
This type of attack has less impact on the global model's MTA, meaning that they are more likely to remain undetected. Although not all classes are equally affected, *IDS backdoors are possible using label-flipping attacks, given a sufficient number of attackers and a well-represented target.* Colluding attackers can realistically create a backdoor that may later be leveraged to evade detection, raising the question of the minimum DPR and MPR necessary for such attacks to be effective. However, the results are dataset-dependent, and some classes remain completely immune to the attack in our experiments, especially in `nb15`.

5.4. Threshold for Effective Attacks

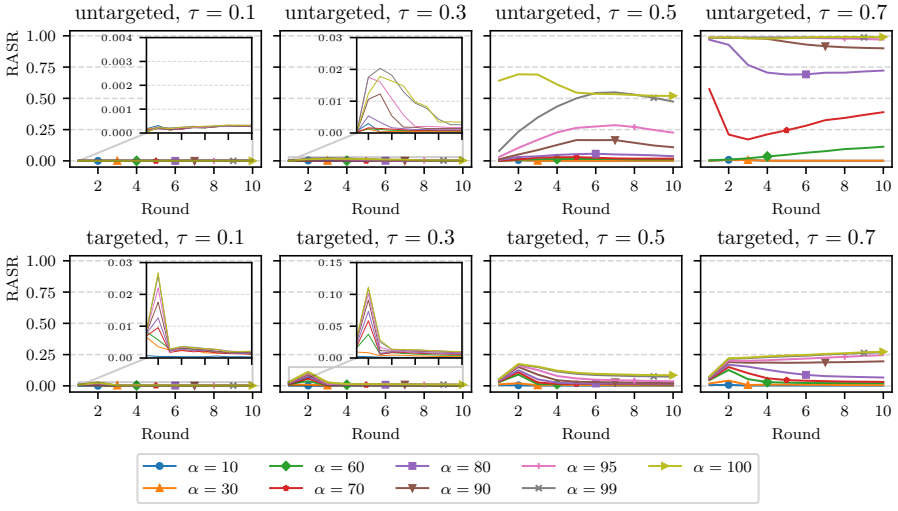
Section 5.3 suggests that the number of attackers is a critical factor in the effectiveness of targeted attacks. This experiment aims to understand the critical threshold where label-flipping attacks begin to impact the global model's accuracy by studying both DPR (α) and MPR (τ). Table 7 summarizes the parameters used for this experiment. Our previous experiments highlighted the impact of the number of attackers on the accuracy of the global model.

Figure 10 presents the RASR of considered label-flipping attacks over time, both for untargeted and targeted attacks. The entire figure emphasizes the importance of the number of attackers in the effectiveness of the attack, as most attacks are not impactful while $\tau \leq 0.3$. In particular, with one single attacker ($\tau = 0.1$), the RASR remains well below 0.1, even with a high poisoning rate ($\alpha = 1.0$). $\tau = 0.5$ represents a tipping point in our tests, with RASR values orders of magnitude higher than with $\tau = 0.3$.

More importantly, we can infer from $\Gamma = \tau \times \alpha$ the total quantity of poisoned data due to our IID partitioning. For untargeted attacks in `cicids`, the RASR exceeds 0.5 for $\Gamma > 0.5$ and approaches 1.0 for $\Gamma > 0.67$. For targeted attacks, the RASR exceeds 0.5 for $\Gamma > 0.49$ and approaches 1.0 for $\Gamma > 0.56$. Untargeted attacks on `nb15` are slightly more effective, with the RASR exceeding 0.5 for $\Gamma > 0.49$ and approaching 1.0 for $\Gamma > 0.63$. However, as highlighted in Section 5.3, targeted attacks on `nb15` are close to ineffective, with the RASR never exceeding



(a) CIC-CSE-IDS2018



(b) UNSW-NB15

Figure 10: Evolution of the RASR of poisoning attacks over time, depending on the local poisoning rate (α), the proportion of attackers (τ), and the type of attack. The x-axis represents the number of rounds. The value for targeted attacks is the mean of the targets, from which we exclude the under-represented and ineffective ones (see Figure 9): "Infiltration" and "Injection" in `cicids`, and "Analysis", "Backdoor", "Shellcode", and "Worms" in `nb15`. The FL round is used as the time unit.

Table 8: Experiment parameters for RQ5

Are existing techniques well-suited to detect label-flipping attacks?	
distribution	10-0, 9-1, 5-5
scenario	continuous-100
target	untargeted
partitioner	iid_drop_1, kmeans_drop_1, iid_drop_2, kmeans_drop_2, iid_keep_1, iid_full, kmeans_full, kmeans_keep_1
dataset	cicids, nb15

0.25, even with $\Gamma = 0.7$, our highest overall poisoning rate. Thus, RASR and the tuple (α, τ) exhibit fairly similar variations, albeit not linear: the higher the DPR and MPR, the higher the RASR. However, poisoning the entire local dataset seems more powerful than instantiating more attackers: $\alpha = 100$ and $\tau = 50$ yield higher RASR than $\alpha = 80$ and $\tau = 70$, although the latter represents more affected data overall.

Answering RQ4 / *Is there a critical threshold where label-flipping attacks begin to impact performance?*

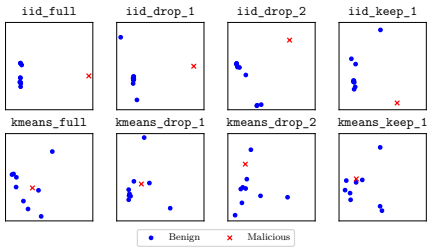
The effectiveness of label-flipping attacks is directly related to the overall quantity of poisoned data. However, this relationship is not linear and there is a critical threshold when α is below 1.0. FL suffers from the same caveat as numerous other distributed systems, where the majority of participants must be honest to ensure the system's security. *However, even in its default configuration and without any defense mechanisms, multiple attackers are necessary to impact the global model's accuracy.*

5.5. Similarity as a Defense Mechanism

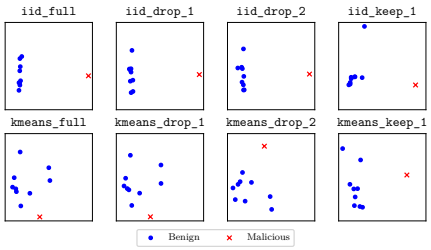
A significant amount of literature has been dedicated to the development of defense mechanisms against poisoning attacks. One of the most represented strategies is the use of similarity metrics to detect poisoned contributions [21, 28, 29, 47]. This relies on the assumption that the attackers' model updates are statistically different from those of benign participants, and that a profile of either of those can be established. For instance, Tolpegin et al. [11] propose using PCA to measure the distance between model updates in the projected space. To assess the effectiveness of this strategy for CIDSs, we explore the impact of different partitioning schemes (*cf*. Section 4.4.1) on the PCA analysis. Table 8 summarizes the parameters used for this experiment.

Figure 11 presents 2D projections of the participants' gradients using PCA for different partitioning schemes. Each point represents a client's contribution to the global model's update in the 10th round. Since `FedAvg` aggregates models by default, we compute the gradients g_i^r as the difference between the participant's model at round r and the last global model, *i.e.*,

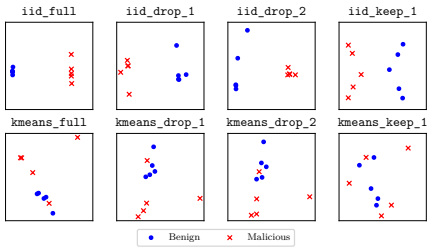
$$g_i^r = \bar{w}^{r-1} - w_i^r. \tag{8}$$



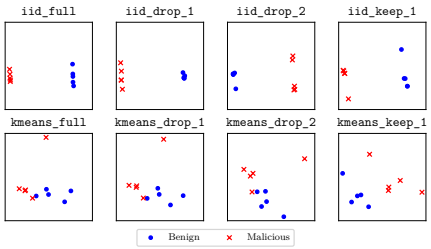
(a) Single attacker on CIC-CSE-IDS2018



(b) Single attacker on UNSW-NB15



(c) Colluding attackers on CIC-CSE-IDS2018



(d) Colluding attackers on UNSW-NB15

Figure 11: 2D projection of gradients using PCA for different partitioning schemes. The top row represents the results for a single attacker (*i.e.*, $\tau = 0.1$), while the bottom row illustrates the results for colluding attackers (*i.e.*, $\tau = 0.5$). Results for the 10th round with `seed=1128` and $\alpha = 1.0$.

This allows us to compare the participants' *direction* rather than their *position* in the model parameter space.

The results in the `iid_full` setting are consistent with the literature: benign participants' contributions are tightly clustered, and the attacker's contributions are easily identifiable as outliers [11]. However, the more heterogeneous partitioning schemes make it less clear, especially with a single attacker. For instance, the `kmeans_keep_1` partitioning scheme on `cicids` (Figure 11a) displays two outliers that are roughly equaly far from the benign participants, one of which is the attacker. In the `kmeans_*` schemes, the attacker is sometimes indistinguishable from the benign participants. Colluding attackers are more easily identifiable, as they are usually grouped together in the less heterogeneous settings. This quality disappears with the `kmeans_*` schemes, especially with `cicids` (Figure 11c). The two datasets exhibit slightly different behaviors, with attackers being more easily identifiable on `nb15` than on `cicids`, depending on the partitioning scheme.

Answering RQ5 / *Are existing techniques well-suited to detect label-flipping attacks?*

The PCA analysis provides an easy and visual way to detect attackers, as long as the data distribution is sufficiently homogeneous. However, this defense mechanism is particularly challenged in more heterogeneous settings. Colluding attackers are more easily identifiable as they will usually form a cluster of their own, highlighting the relevance of mitigation strategies such as `FoolsGold` [21] and `CONTRA` [29] which detect colluding attackers in heterogeneous environments using their similarity. *Overall, similarity-based defense mechanisms are effective in detecting attackers in homogeneous environments, but their effectiveness decreases with the heterogeneity of the data distribution.*

6. Discussion

6.1. Main takeaways

Attack Effectiveness. Label-flipping attacks can significantly degrade the performance of FL models for IDSs, their effectiveness being directly related to the total quantity of poisoned data. In IID settings, this can be approximated by the product of DPR (α) and MPR (τ), but this relationship is distribution-dependent. However, these attacks are not very practical because of the high number of malicious participants required for significant impact. They can, nonetheless, effectively model various data quality issues in training, such as mislabeled events (*e.g.*, an entire event missed by the operator) or noisy data (*e.g.*, multiple labels for similar samples).

IDS Evasion with Targeted Attacks. Targeted attacks are particularly effective against well-detected targets, although their success can be mitigated by the

model's generalization capabilities (Section 5.3). This slightly contradicts the observations made by Tolpegin et al. [11], which can be explained by the class overlap inherent to IDS use cases. For more effective evasion techniques, ML backdoors, as reviewed by Merzouk et al. [24], may be more suitable.

Implications for Defense and Operation. While label-flipping attacks can pose a threat to FL-based IDSs, they require significant resources to be effective, suggesting some inherent resilience. However, proper monitoring and defense mechanisms remain essential, especially in scenarios where data quality cannot be guaranteed. Gradient similarity analysis can be effective for detection (see Section 2.3), although its reliability decreases in heterogeneous settings (Section 5.5). Mitigation strategies must also be carefully tailored to account for the dynamic behavior of attacks, particularly in constrained environments such as IoT (Section 5.2).

6.2. Limitations of this study

Systematically reviewing the behavior of such complex systems involves a significant number of variables. As a result, the number of experiments required to cover all possible scenarios is impractical. We made critical choices in the evaluation, and discuss bellow the caveats associated with them.

Scalability. The experiments carried out in this study are limited to a small number of participants (*i.e.*, 10) and do not account for variations in this number. While this setting is representative of real-world collaborative use cases, a typical cross-silo federation between organizations, its scalability can be questioned. However, some related studies [24, 35] have shown that the impact of poisoning attacks is not significantly influenced by the number of participants, as long as the proportions of malicious participants are preserved. We thus expect these experiments to act similarly with a larger number of participants.

Heterogeneity Simulation. While RQ5 touches on the topic of data distribution, it focuses on a caricatural depiction that is not fully representative of a real-world setting. In fact, all evaluations that rely on partitioning datasets among participants are limited by design in their representativeness, notably due to the dependencies between training samples. Although we can expect the results of Section 5.5 to be applicable to other data distributions, this also emphasizes the need for independent datasets tailored for distributed intrusion detection.

Generalizability of the results. More generally, the differences observed between the results on the two considered datasets (*i.e.*, UNSW-NB15 and CSE-CIC-IDS2018) raise the question of whether the results observed here can be generalized to other datasets and real-world deployments. While correlations can be observed with existing work [23, 36], including on other datasets with NSL-KDD [23], the lack of a common benchmark for federated intrusion detection systems makes it difficult to compare the results of different studies. This calls for extended evaluations of other datasets and models, as well as the development of a common benchmark for distributed IDSs.

7. Conclusion and Takeaways

The literature on the impact of poisoning attacks on FL that specifically cover intrusion detection use cases is scarce, and in it, label-flipping attacks have been overlooked. This paper fills this gap by providing a comprehensive analysis of the impact of label-flipping attacks on FL for IDSs. We evaluated the impact of untargeted and targeted label-flipping attacks on the performance of FL models trained on CSE-CIC-IDS2018 and UNSW-NB15 using a standardized feature set to allow extension of this work.

Our results highlight that (i) label-flipping attacks can have a significant impact on the performance of FL models, but their success rate is tied to the number of flipped samples overall; (ii) IDS backdoors using label-flipping attacks can be significantly mitigated by the model's generalization capabilities; (iii) mitigation strategies must be adapted to the use case specificities (*e.g.*, constrained environments); (iv) gradient similarity *can* be used to detect label-flipping attacks, but its effectiveness is challenged in heterogeneous settings.

However, many open questions remain to be addressed. First, while the comparison with existing works seems to corroborate our results, this study calls to be extended to other datasets, feature sets, model architectures, and FL aggregation strategies, but also to larger-scale use cases. Second, the provided evaluation framework can be used to evaluate the efficiency of existing countermeasures or to develop new ones. Consequently, evaluating existing defenses in these settings is a critical next step of this work. Finally, this study also highlights community-wide limitations in evaluating FL models for IDSs, notably the lack of appropriate datasets and a common benchmark. We strongly believe that this work is a first step in this direction and advocate for better evaluation and comparison of FL models and aggregation strategies in the context of intrusion detection.

References

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, B. A. y Arcas, Communication-efficient learning of deep networks from decentralized data, in: A. Singh, J. Zhu (Eds.), Proceedings of the 20th International Conference on Artificial Intelligence and Statistics, volume 54 of *Proceedings of Machine Learning Research*, PMLR, 2017-04-20/2017-04-22, pp. 1273–1282.

[2] T. D. Nguyen, S. Marchal, M. Miettinen, H. Fereidooni, N. Asokan, A.-R. Sadeghi, DÏoT: A Federated Self-learning Anomaly Detection System for IoT, in: 2019 IEEE 39th International Conference on Distributed Computing Systems (ICDCS), volume 2019-July, IEEE, 2019, pp. 756–767. doi:10.1109/ICDCS.2019.00080.

[3] H. Liu, S. Zhang, P. Zhang, X. Zhou, X. Shao, G. Pu, Y. Zhang, Blockchain and Federated Learning for Collaborative Intrusion Detection in Vehicular

Edge Computing, IEEE Trans. Veh. Technol. 70 (2021) 6073–6084. doi:10.1109/TVT.2021.3076780.

[4] S. Agrawal, S. Sarkar, O. Aouedi, G. Yenduri, K. Piamrat, M. Alazab, S. Bhattacharya, P. K. R. Maddikunta, T. R. Gadekallu, Federated Learning for intrusion detection system: Concepts, challenges and future directions, Computer Communications 195 (2022) 346–361. doi:10.1016/j.comcom.2022.09.012.

[5] L. Lavaur, M.-O. Pahl, Y. Busnel, F. Autrel, The Evolution of Federated Learning-based Intrusion Detection and Mitigation: A Survey, TNSM (2022).

[6] M. Alazab, S. P. R M, P. M, P. Reddy, T. R. Gadekallu, Q.-V. Pham, Federated Learning for Cybersecurity: Concepts, Challenges and Future Directions, IEEE Trans. Ind. Inf. (2021) 1–1. doi:10/gnm4dj.

[7] E. Fedorchenko, E. Novikova, A. Shulepov, Comparative Review of the Intrusion Detection Systems Based on Federated Learning: Advantages and Open Challenges, Algorithms 15 (2022) 247. doi:10.3390/a15070247.

[8] B. Ghimire, D. B. Rawat, Recent Advances on Federated Learning for Cybersecurity and Cybersecurity for Federated Learning for Internet of Things, IEEE Internet of Things Journal 9 (2022) 8229–8249. doi:10.1109/JIOT.2022.3150363.

[9] E. M. Campos, P. F. Saura, A. González-Vidal, J. L. Hernández-Ramos, J. B. Bernabé, G. Baldini, A. Skarmeta, Evaluating Federated Learning for intrusion detection in Internet of Things: Review and challenges, Computer Networks 203 (2022) 108661. doi:10.1016/j.comnet.2021.108661.

[10] A. N. Bhagoji, S. Chakraborty, P. Mittal, S. Calo, Analyzing Federated Learning through an Adversarial Lens, in: Proceedings of the 36th International Conference on Machine Learning, PMLR, 2019, pp. 634–643.

[11] V. Tolpegin, S. Truex, M. E. Gursoy, L. Liu, Data Poisoning Attacks Against Federated Learning Systems, in: L. Chen, N. Li, K. Liang, S. Schneider (Eds.), Computer Security – ESORICS 2020, Lecture Notes in Computer Science, Springer International Publishing, Cham, 2020, pp. 480–501. doi:10.1007/978-3-030-58951-6_24.

[12] D. J. Beutel, T. Topal, A. Mathur, X. Qiu, T. Parcollet, N. D. Lane, Flower: A friendly federated learning research framework, arXiv preprint arXiv:2007.14390 (2020). arXiv:2007.14390.

[13] O. Yadan, Hydra - A framework for elegantly configuring complex applications, Github, 2019.

[14] L. Lavaur, Y. Busnel, F. Autrel, Systematic analysis of label-flipping attacks against federated learning in collaborative intrusion detection systems, in: Proceedings of the 19th International Conference on Availability, Reliability and Security (ARES), Vienna, Austria, 2024.

[15] P. Kairouz, H. B. McMahan, B. Avent, A. Bellet, M. Bennis, A. N. Bhagoji, K. Bonawitz, Z. Charles, G. Cormode, R. Cummings, R. G. L. D'Oliveira, H. Eichner, S. E. Rouayheb, D. Evans, J. Gardner, Z. Garrett, A. Gascón, B. Ghazi, P. B. Gibbons, M. Gruteser, Z. Harchaoui, C. He, L. He, Z. Huo, B. Hutchinson, J. Hsu, M. Jaggi, T. Javidi, G. Joshi, M. Khodak, J. Konečný, A. Korolova, F. Koushanfar, S. Koyejo, T. Lepoint, Y. Liu, P. Mittal, M. Mohri, R. Nock, A. Özgür, R. Pagh, M. Raykova, H. Qi, D. Ramage, R. Raskar, D. Song, W. Song, S. U. Stich, Z. Sun, A. T. Suresh, F. Tramèr, P. Vepakomma, J. Wang, L. Xiong, Z. Xu, Q. Yang, F. X. Yu, H. Yu, S. Zhao, Advances and Open Problems in Federated Learning, arXiv:1912.04977 [cs, stat] (2021). arXiv:1912.04977.

[16] Y. Huang, L. Chu, Z. Zhou, L. Wang, J. Liu, J. Pei, Y. Zhang, Personalized Cross-Silo Federated Learning on Non-IID Data, AAAI 35 (2021) 7865–7873. doi:10.1609/aaai.v35i9.16960.

[17] L. Lavaur, Y. Busnel, F. Autrel, Demo: Highlighting the limits of federated learning in intrusion detection, in: Proceedings of the 44th International Conference on Distributed Computing Systems (ICDCS), Jersey City, NJ, USA, 2024.

[18] N. Rodríguez-Barroso, D. Jiménez-López, M. V. Luzón, F. Herrera, E. Martínez-Cámara, Survey on federated learning threats: Concepts, taxonomy on attacks and defences, experimental study and challenges, Information Fusion 90 (2023) 148–173. doi:10.1016/j.inffus.2022.09.011.

[19] L. Lamport, R. Shostak, M. Pease, The Byzantine generals problem, in: Concurrency: The Works of Leslie Lamport, Association for Computing Machinery, New York, NY, USA, 1982, pp. 203–226.

[20] J. R. Douceur, The Sybil Attack, in: P. Druschel, F. Kaashoek, A. Rowstron (Eds.), Peer-to-Peer Systems, Lecture Notes in Computer Science, Springer, Berlin, Heidelberg, 2002, pp. 251–260. doi:10.1007/3-540-45748-8_24.

[21] C. Fung, C. J. M. Yoon, I. Beschastnikh, The Limitations of Federated Learning in Sybil Settings, in: 23rd International Symposium on Research in Attacks, Intrusions and Defenses ({RAID} 2020), {USENIX} Association, San Sebastian, 2020, pp. 301–316.

[22] M. Fang, X. Cao, J. Jia, N. Gong, Local Model Poisoning Attacks to Byzantine-Robust Federated Learning, in: 29th USENIX Security Symposium (USENIX Security 20), 2020, pp. 1605–1622.

[23] Y. Zhang, Y. Zhang, Z. Zhang, H. Bai, T. Zhong, M. Song, Evaluation of data poisoning attacks on federated learning-based network intrusion detection system, in: 2022 IEEE 24th Int Conf on High Performance Computing & Communications; 8th Int Conf on Data Science & Systems; 20th Int Conf on Smart City; 8th Int Conf on Dependability in Sensor, Cloud & Big Data Systems & Application (HPCC/DSS/SmartCity/DependSys), 2022, pp. 2235–2242. doi:10.1109/HPCC-DSS-SmartCity-DependSys57074.2022.00330.

[24] M. A. Merzouk, F. Cuppens, N. Boulahia-Cuppens, R. Yaich, Parameterizing poisoning attacks in federated learning-based intrusion detection, in: Proceedings of the 18th International Conference on Availability, Reliability and Security, ARES '23, Association for Computing Machinery, New York, NY, USA, 2023, pp. 1–8. doi:10.1145/3600160.3605090.

[25] P. Blanchard, E. M. El Mhamdi, R. Guerraoui, J. Stainer, Machine learning with adversaries: Byzantine tolerant gradient descent, Advances in Neural Information Processing Systems 30 (2017).

[26] D. Yin, Y. Chen, R. Kannan, P. Bartlett, Byzantine-Robust Distributed Learning: Towards Optimal Statistical Rates, in: Proceedings of the 35th International Conference on Machine Learning, PMLR, 2018, pp. 5650–5659.

[27] S. Shen, S. Tople, P. Saxena, Auror: Defending against poisoning attacks in collaborative deep learning systems, in: Proceedings of the 32nd Annual Conference on Computer Security Applications, ACM, New York, NY, USA, 2016, pp. 508–519. doi:10.1145/2991079.2991125.

[28] T. D. Nguyen, P. Rieger, H. Chen, H. Yalame, H. Möllering, H. Fereidooni, S. Marchal, M. Miettinen, A. Mirhoseini, S. Zeitouni, F. Koushanfar, A.-R. Sadeghi, T. Schneider, FLAME: Taming Backdoors in Federated Learning, in: 31st USENIX Security Symposium (USENIX Security 22), 2022, pp. 1415–1432.

[29] S. Awan, B. Luo, F. Li, CONTRA: Defending Against Poisoning Attacks in Federated Learning, in: E. Bertino, H. Shulman, M. Waidner (Eds.), Computer Security – ESORICS 2021, Lecture Notes in Computer Science, Springer International Publishing, Cham, 2021, pp. 455–475. doi:10.1007/978-3-030-88418-5_22.

[30] L. Zhao, S. Hu, Q. Wang, J. Jiang, C. Shen, X. Luo, P. Hu, Shielding Collaborative Learning: Mitigating Poisoning Attacks through Client-Side Detection, 2020. arXiv:1910.13111.

[31] L. Lavaur, P.-M. Lechevalier, Y. Busnel, R. Ludinard, G. Texier, M.-O. Pahl, RADAR: Model quality assessment for reputation-aware collaborative federated learning, in: Proceedings of the 43rd International Symposium on Reliable Distributed Systems (SRDS), Charlotte, NC, USA, 2024-09, 2024.

[32] F. Nuding, R. Mayer, Data Poisoning in Sequential and Parallel Federated Learning, in: Proceedings of the 2022 ACM on International Workshop on Security and Privacy Analytics, ACM, Baltimore MD USA, 2022, pp. 24–34. doi:10.1145/3510548.3519372.

[33] G. Sun, Y. Cong, J. Dong, Q. Wang, L. Lyu, J. Liu, Data Poisoning Attacks on Federated Machine Learning, IEEE Internet of Things Journal 9 (2022) 11365–11375. doi:10.1109/JIOT.2021.3128646.

[34] T. D. Nguyen, P. Rieger, M. Miettinen, A.-R. Sadeghi, Poisoning Attacks on Federated Learning-based IoT Intrusion Detection System, in: Proceedings 2020 Workshop on Decentralized IoT Systems and Security, Internet Society, San Diego, CA, 2020. doi:10.14722/diss.2020.23003.

[35] N. C. Vy, N. H. Quyen, P. T. Duy, V.-H. Pham, Federated Learning-Based Intrusion Detection in the Context of IIoT Networks: Poisoning Attack and Defense, in: M. Yang, C. Chen, Y. Liu (Eds.), Network and System Security, volume 13041, Springer International Publishing, Cham, 2021, pp. 131–147. doi:10.1007/978-3-030-92708-0_8.

[36] R. Yang, H. He, Y. Wang, Y. Qu, W. Zhang, Dependable federated learning for IoT intrusion detection against poisoning attacks, Computers & Security 132 (2023) 103381. doi:10.1016/j.cose.2023.103381.

[37] M. Tavallaee, E. Bagheri, W. Lu, A. A. Ghorbani, A detailed analysis of the KDD CUP 99 data set, in: 2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications, Cisda, IEEE, 2009, pp. 1–6. doi:10.1109/CISDA.2009.5356528.

[38] N. Moustafa, J. Slay, UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set), in: 2015 Military Communications and Information Systems Conference (MilCIS), 2015, pp. 1–6. doi:10.1109/MilCIS.2015.7348942.

[39] ACM, Artifact Review and Badging v1.1, https://www.acm.org/publications/policies/artifact-review-and-badging-current, 2020.

[40] M. Sarhan, S. Layeghy, M. Portmann, Towards a Standard Feature Set for Network Intrusion Detection System Datasets, Mobile Netw Appl 27 (2022) 357–370. doi:10.1007/s11036-021-01843-0.

[41] S. Layeghy, M. Portmann, On Generalisability of Machine Learning-based Network Intrusion Detection Systems, 2022. arXiv:2205.04112.

[42] M. Sarhan, S. Layeghy, M. Portmann, Towards a Standard Feature Set for Network Intrusion Detection System Datasets, 2021. arXiv:2101.11315.

[43] ntop, nprobe - an extensible netflow v5/v9/ipfix probe for ipv4/v6, ???? URL: https://www.ntop.org/products/netflow/nprobe/.

[44] I. Sharafaldin, A. Habibi Lashkari, A. A. Ghorbani, Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization, in: Proceedings of the 4th International Conference on Information Systems Security and Privacy, SCITEPRESS - Science and Technology Publications, Funchal, Madeira, Portugal, 2018, pp. 108–116. doi:10.5220/0006639801080116.

[45] S. I. Popoola, G. Gui, B. Adebisi, M. Hammoudeh, H. Gacanin, Federated Deep Learning for Collaborative Intrusion Detection in Heterogeneous Networks, in: 2021 IEEE 94th Vehicular Technology Conference (VTC2021-Fall), 2021, pp. 1–6. doi:10.1109/VTC2021-Fall52928.2021.9625505.

[46] J. B. W. Webber, A bi-symmetric log transformation for wide-range data, Meas. Sci. Technol. 24 (2012) 027001. doi:10.1088/0957-0233/24/2/027001.

[47] X. Cao, M. Fang, J. Liu, N. Z. Gong, FLTrust: Byzantine-robust Federated Learning via Trust Bootstrapping, 2022. doi:10.48550/arXiv.2012.13995. arXiv:2012.13995.